Trend Rotation and Multi-Asset Momentum

Two systematic strategies compared: SPY/IEF moving-average rotation and a six-ETF monthly momentum rotation

Prepared October 19, 2026

Hypothetical backtested results. For discussion purposes only. Generated by an AI multi-agent research pipeline.

Executive Summary

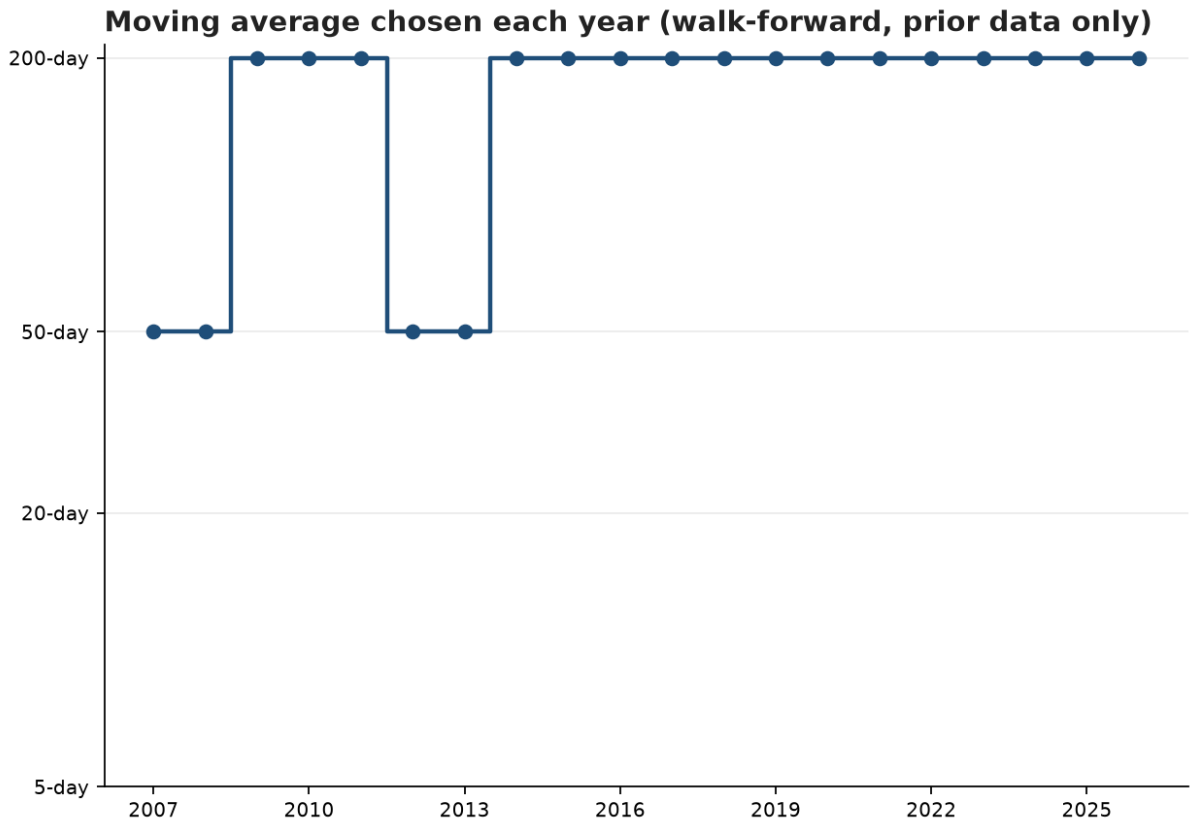
- Return since 2011: SPY's CAGR (14.1%) exceeded every momentum lookback (7.3% to 11.8%, before costs) and the MA strategy (9.3%, net of costs). QQQ was highest at 18.7%.
- Sharpe since 2011: 6-month momentum (0.79) was in line with SPY (0.77); QQQ was highest (0.86). The MA strategy (0.67) and the 1, 3, 12 and 24-month lookbacks (0.52 to 0.71) were below SPY.
- Drawdown since 2011: momentum max drawdowns ranged from -20.8% (6-month) to -28.8% (24-month) and the MA strategy's was -26.4%, all shallower than SPY (-33.7%) and QQQ (-35.1%).
- Verdict: 6-month momentum was the strongest variant on return, Sharpe and drawdown; SPY and QQQ still beat it on return and QQQ on Sharpe. Both approaches are drawdown-control tools, not return maximisers.
- Since 2007, a 60/40 mix beat the MA strategy on CAGR (8.4% vs 7.6%) and Sharpe (0.63 vs 0.53), and SPY's Sharpe was slightly higher (0.55 vs 0.53). The MA strategy's max drawdown was -26.4% vs SPY -55.2%.

The Problem: Deep Equity Drawdowns

- Since 2007, SPY compounded at 10.9% a year, but its maximum drawdown was -55.2% and its worst rolling 1-year return was -47.4%.
- In 2008 SPY returned -36.8%; in 2022 it returned -18.2%. Losses of this size can force investors to sell at the wrong time.
- Both strategies in this deck aim to reduce exposure to sustained equity declines using simple, rules-based trend signals.
- The trade-off is explicit: lower drawdowns have come with lower long-run returns than SPY, as the following slides show.

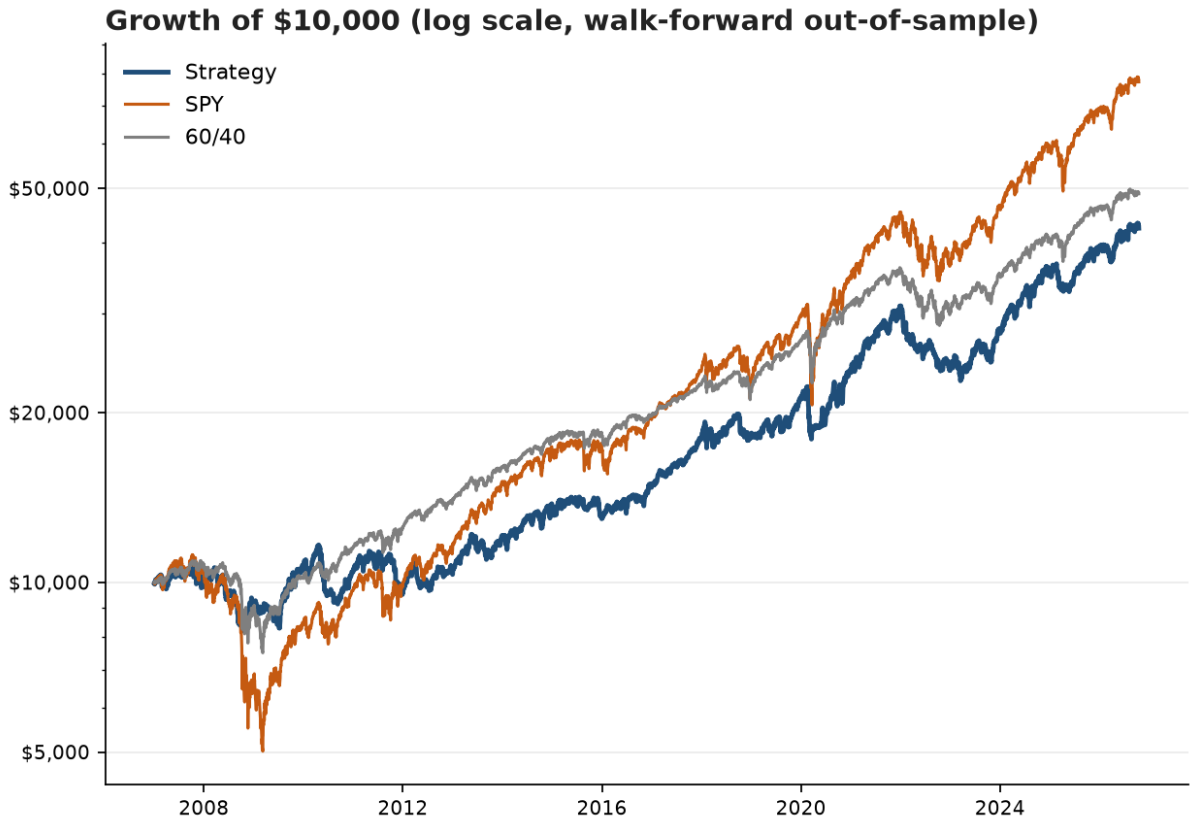
How the MA Trend Strategy Works: Rules and Walk-Forward Choice

- A tactical SPY/Treasury allocation: hold SPY when SPY is above its moving average, otherwise hold IEF (intermediate Treasuries).
- Candidate moving averages: 5, 20, 50 or 200 days. Each January the average is re-selected using only data available before that date.
- Over 20 years of walk-forward selection, the 200-day average was chosen 16 times and the 50-day 4 times (chart).
- Results are net of modeled trading costs. Since 2007 the strategy switched 158 times and held SPY 77.0% of the time.
- Current allocation: 100.0% SPY.



MA Strategy Growth vs SPY Since 2007

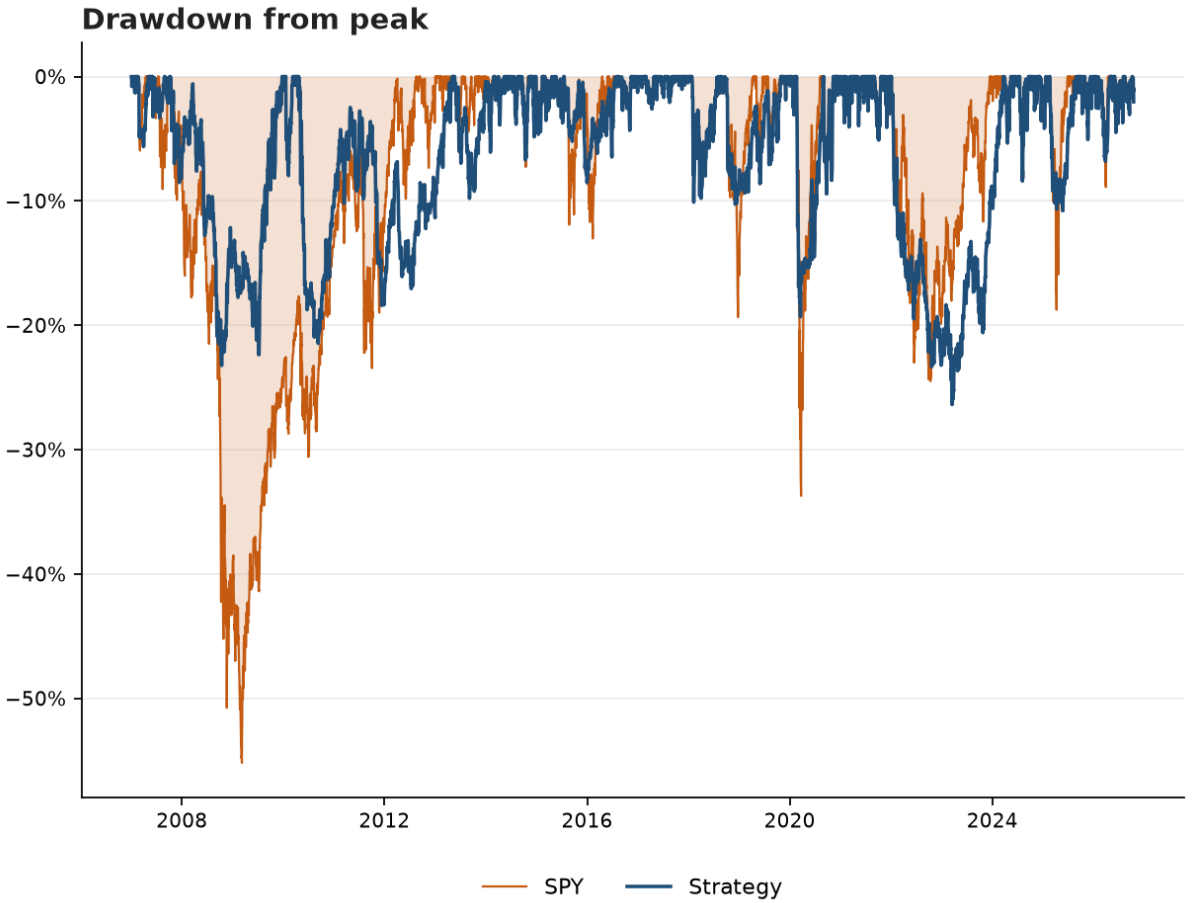
- $10,000 invested in January 2007 grew to $42,495 in the MA strategy vs $77,253 in SPY and $48,905 in a 60/40 mix.
- CAGR since 2007: 7.6% for the strategy vs 10.9% for SPY and 8.4% for 60/40.
- Volatility was lower than SPY's (12.4% vs 19.5%), but Sharpe was slightly below SPY's (0.53 vs 0.55) and 60/40's (0.63).
- The strategy captured 54.9% of SPY's upside and 53.5% of its downside: it participates in only part of rising markets.



Hypothetical backtested performance. Not actual trading; not net of fees. MA strategy results are net of modeled trading costs; momentum results exclude trading costs. Past performance does not guarantee future results. See Important Disclosures.

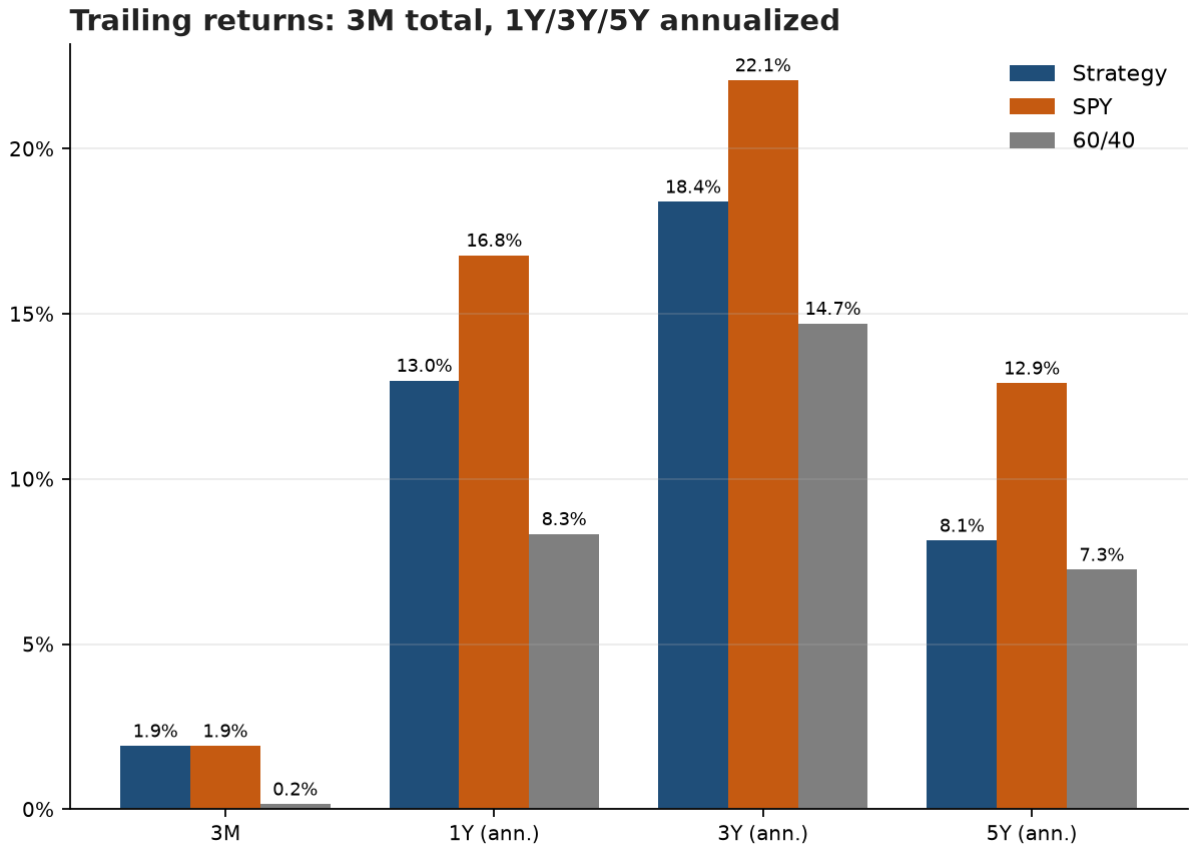
Risk and Drawdown Protection vs SPY

- Maximum drawdown since 2007: -26.4% for the strategy vs -55.2% for SPY and -31.4% for 60/40.
- Worst rolling 1-year return: -23.1% vs -47.4% for SPY. Best rolling 1-year return: 44.6% vs 77.5% for SPY.
- In 2008 the strategy returned -6.8% vs SPY's -36.8%, as the signal moved into Treasuries.
- Protection is not uniform: in 2022 stocks and bonds fell together and the strategy returned -22.7% vs SPY's -18.2%.
- Calmar ratio since 2007: 0.29 vs 0.20 for SPY.



MA Strategy Performance by Period: 3M, 1Y, 3Y, 5Y

- 3M: 1.9% for both the strategy and SPY; the strategy held SPY throughout, so results were identical.
- 1Y: 13.0% vs SPY 16.8%; max drawdown -6.8% vs -8.9%, but Sharpe was lower (0.78 vs 1.00).
- 3Y (annualized): 18.4% vs SPY 22.1%; Sharpe in line (1.11 vs 1.10) with a shallower max drawdown (-10.8% vs -18.8%).
- 5Y (annualized): 8.1% vs SPY 12.9%; Sharpe 0.39 vs 0.57 and a deeper max drawdown (-26.4% vs -24.5%), driven by 2022.
- Drawdown benefits have depended on the period: they were clear over 3Y and since 2007, absent over 5Y.



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Risk and Return by Period: Strategy vs SPY

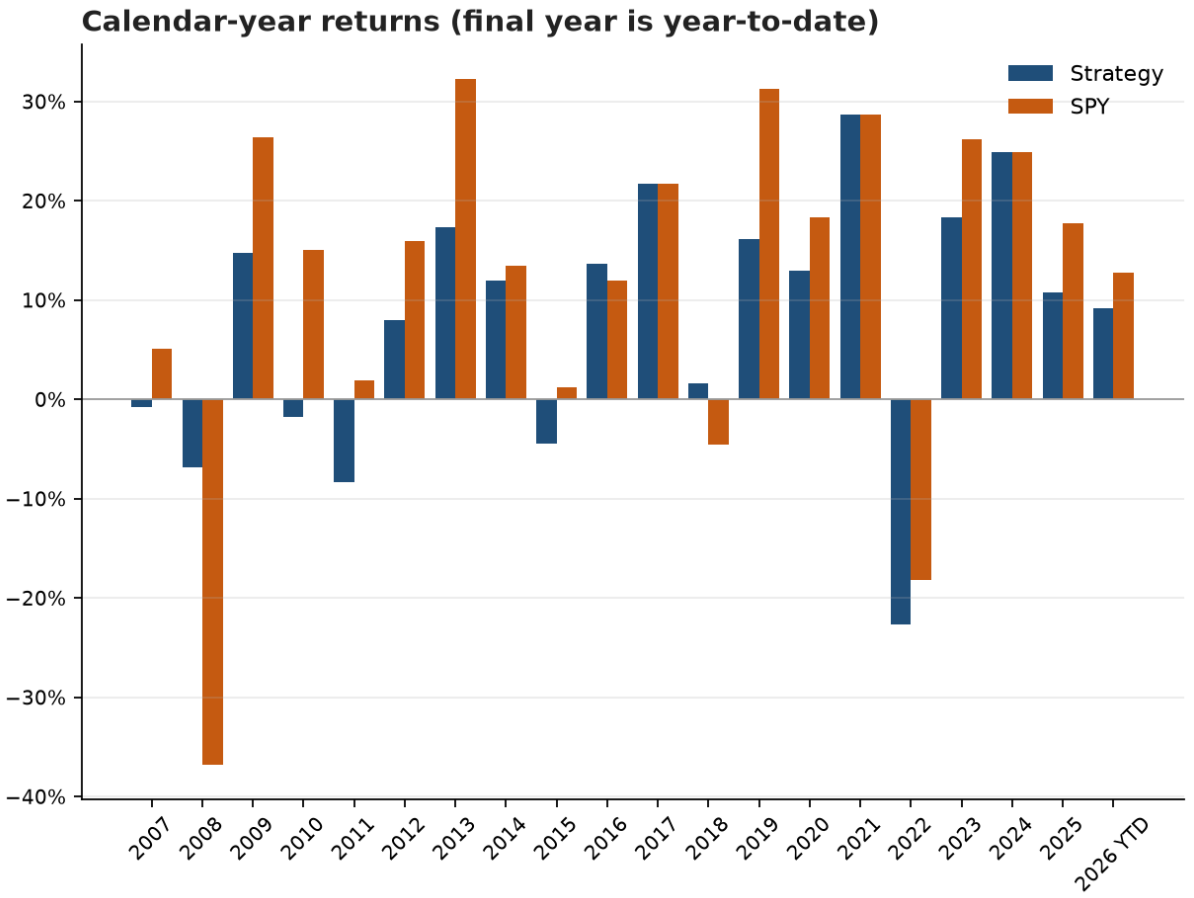
Metric	3M Strat.	3M SPY	1Y Strat.	1Y SPY	3Y Strat.	3Y SPY	5Y Strat.	5Y SPY	SI Strat.	SI SPY
Return (3M total, else annualized)	**1.9%**	**1.9%**	**13.0%**	**16.8%**	**18.4%**	**22.1%**	**8.1%**	**12.9%**	**7.6%**	**10.9%**
Total return (cumulative)	1.9%	1.9%	13.0%	16.8%	65.6%	81.5%	47.6%	83.0%	324.9%	672.5%
Volatility (ann.)	11.1%	11.1%	11.9%	12.7%	12.0%	15.3%	12.3%	17.1%	12.4%	19.5%
Sharpe ratio	n/a	n/a	0.78	1.00	1.11	1.10	0.39	0.57	0.53	0.55
Sortino ratio	n/a	n/a	1.11	1.47	1.57	1.63	0.54	0.83	0.72	0.77
Max drawdown	-3.1%	-3.1%	-6.8%	-8.9%	-10.8%	-18.8%	-26.4%	-24.5%	-26.4%	-55.2%
Calmar ratio	n/a	n/a	1.90	1.89	1.70	1.18	0.31	0.53	0.29	0.20
Beta vs SPY	1.00	1.00	0.87	1.00	0.58	1.00	0.46	1.00	0.29	1.00
Correlation vs SPY	1.00	1.00	0.94	1.00	0.74	1.00	0.65	1.00	0.45	1.00
Up capture	100.0%	100.0%	91.7%	100.0%	83.3%	100.0%	65.7%	100.0%	54.9%	100.0%
Down capture	100.0%	100.0%	94.6%	100.0%	83.3%	100.0%	65.9%	100.0%	53.5%	100.0%
% days positive	44.6%	44.6%	52.6%	52.6%	56.2%	56.4%	53.6%	53.7%	54.1%	55.0%
Best day	1.8%	1.8%	1.9%	2.9%	2.5%	10.5%	2.5%	10.5%	4.4%	14.5%
Worst day	-1.5%	-1.5%	-2.6%	-2.6%	-3.0%	-5.9%	-3.0%	-5.9%	-5.8%	-10.9%
Switches / % time in SPY	0 / 100.0%		2 / 95.2%		8 / 92.2%		30 / 78.2%		158 / 77.0%	

60/40 benchmark	3M	1Y	3Y	5Y	SI
Return (3M total, else ann.)	0.2%	8.3%	14.7%	7.3%	8.4%
Total return (cumulative)	0.2%	8.3%	50.7%	41.7%	389.0%
Volatility (ann.)	7.8%	8.5%	9.7%	11.0%	11.2%
Sharpe ratio	n/a	0.55	1.02	0.35	0.63
Max drawdown	-2.8%	-6.0%	-10.6%	-21.0%	-31.4%
Calmar ratio	n/a	1.39	1.39	0.35	0.27

Periods end on the last completed session. 3M: 2026-07-17 to 2026-10-16, 1Y: 2025-10-17 to 2026-10-16, 3Y: 2023-10-17 to 2026-10-16, 5Y: 2021-10-18 to 2026-10-16, SI: 2007-01-03 to 2026-10-16. SI = since the walk-forward start. Ratios use the 13-week T-bill as the risk-free rate. Current allocation: 100.0% SPY.

MA Strategy Calendar-Year Returns vs SPY and 60/40

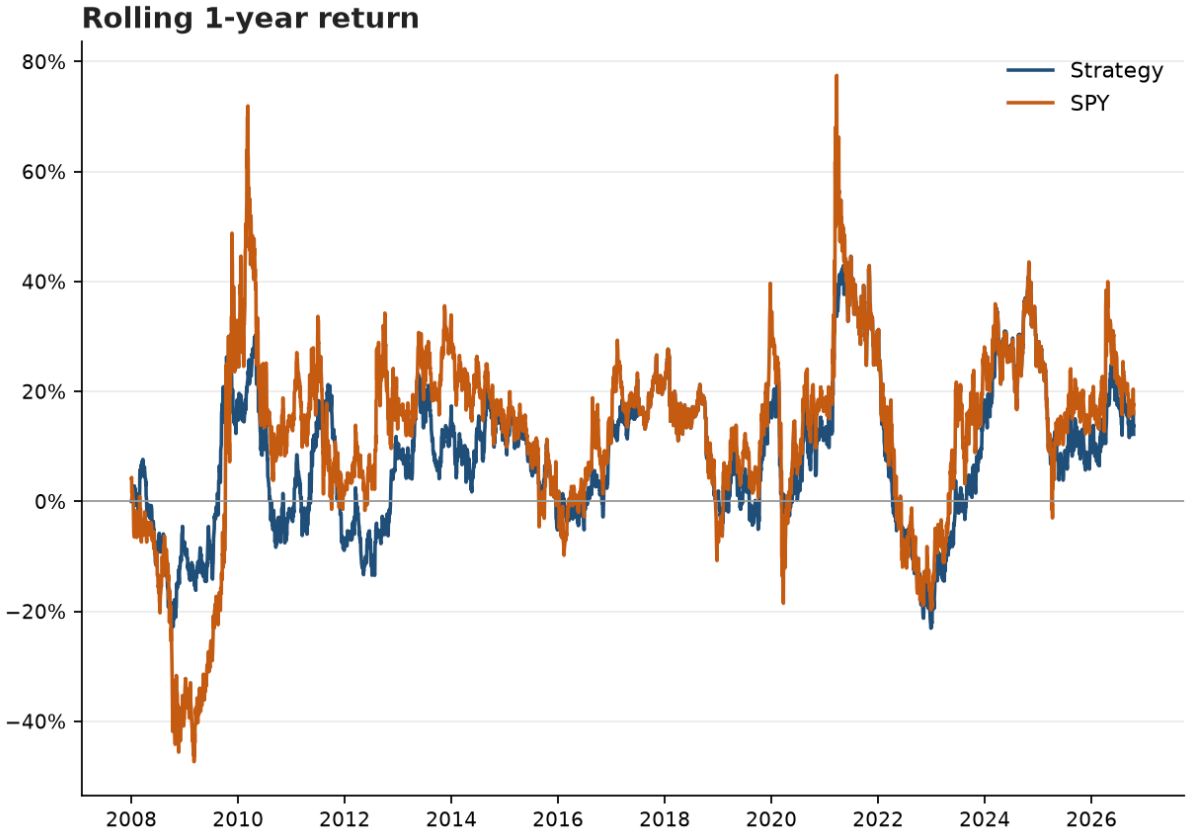
- Strongest relative years came in equity sell-offs: 2008 (-6.8% vs SPY -36.8%) and 2018 (1.6% vs SPY -4.6%).
- In strong, steady markets it often matched SPY when fully invested: 2017 (21.7%), 2021 (28.7%) and 2024 (24.9%) for both.
- It lagged in choppy or V-shaped years: 2011 (-8.3% vs 1.9%), 2019 (16.2% vs 31.2%) and 2020 (12.9% vs 18.3%).
- 2026 year-to-date (through 2026-10-16): 9.1% vs SPY 12.8% and 60/40 6.3%.



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When the MA Strategy Lags: Honest Trade-Offs

- 1Y: Sharpe was below SPY's (0.78 vs 1.00), with a lower return (13.0% vs 16.8%).
- 5Y: Sharpe was below SPY's (0.39 vs 0.57) and max drawdown was deeper (-26.4% vs -24.5%).
- Since 2007 (SI): Sharpe was slightly below SPY's (0.53 vs 0.55); 60/40 beat it on CAGR (8.4% vs 7.6%) and Sharpe (0.63 vs 0.53).
- It beat SPY in only 18.7% of rolling 1-year periods; the median rolling 1-year excess return was -4.1%.
- Whipsaws cost returns in trendless markets, and the IEF leg does not help when bonds and stocks fall together, as in 2022.



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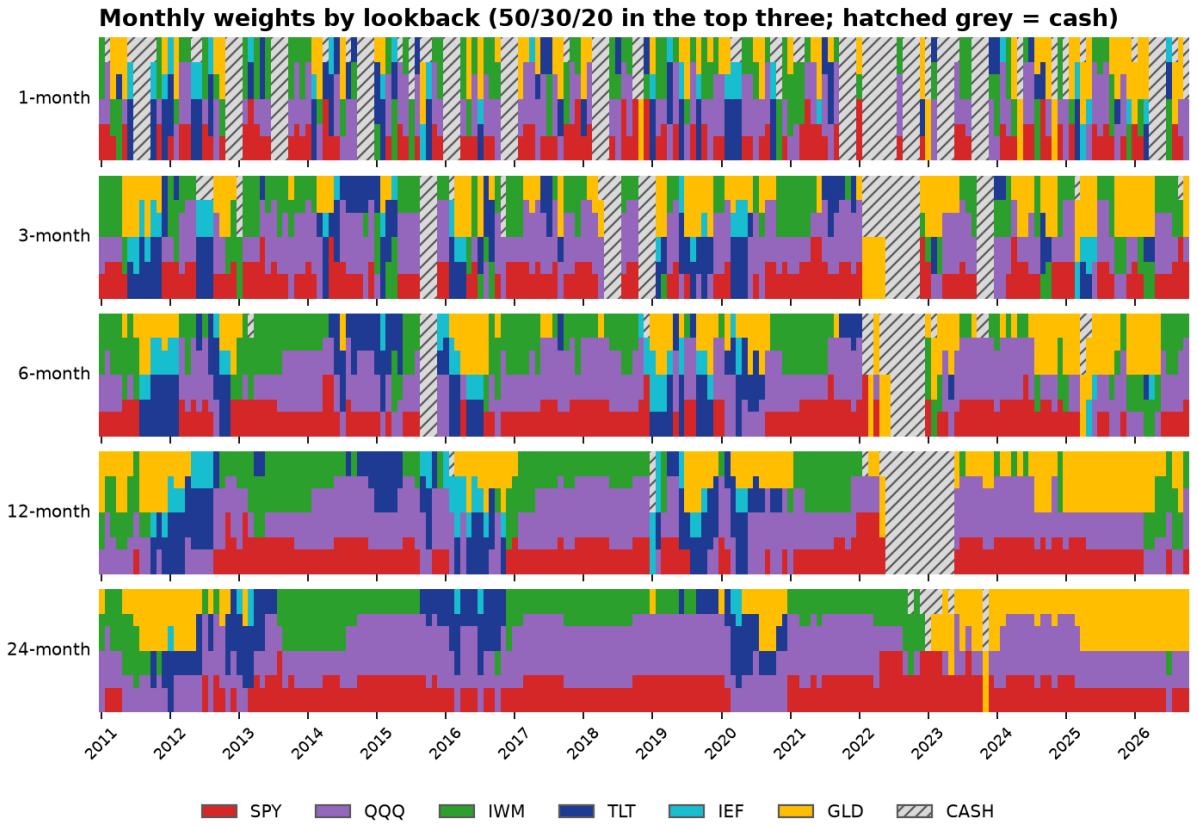
Context: Each Moving Average Over the Full Period (IN-SAMPLE)

Rule (IN-SAMPLE)	CAGR	Volatility	Sharpe	Max DD	Switches	% in SPY	Years chosen
5-day MA switch	-3.5%	12.9%	-0.33	-66.5%	1330	59.4%	0
20-day MA switch	4.0%	12.3%	0.26	-37.1%	570	66.3%	0
50-day MA switch	6.0%	12.0%	0.42	-29.8%	348	70.2%	4
200-day MA switch	9.4%	12.5%	0.66	-26.4%	112	79.1%	16

IN-SAMPLE (2007-01-03 to 2026-10-16): these rows use hindsight. Picking the best row after the fact would overstate results. The strategy instead picks a moving average each January using only earlier data, and that walk-forward result is what the performance pages show. "Years chosen" counts how many years the walk-forward selected each rule.

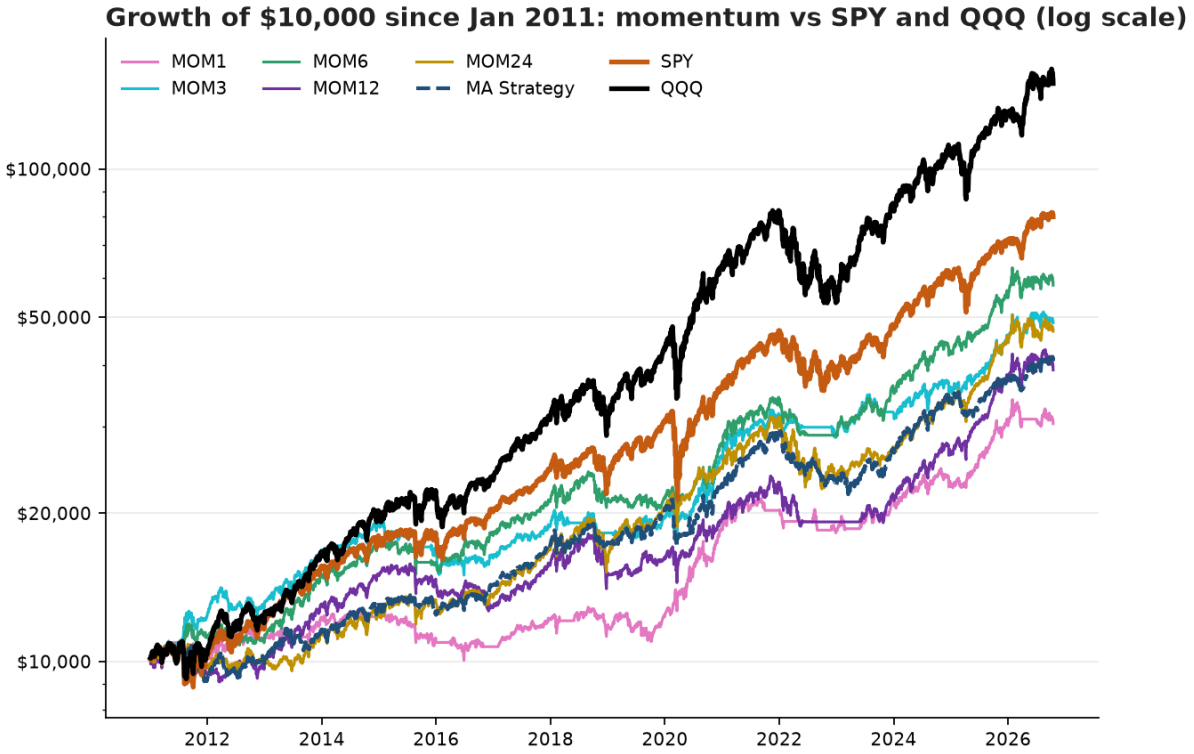
How Multi-Asset Momentum Works: Monthly 50/30/20 Allocation

- Five sub-strategies with 1, 3, 6, 12 and 24-month lookbacks, run from January 2011 and rebalanced monthly.
- At each month-end, each ranks SPY, QQQ, TLT, GLD, IWM and IEF by trailing return and holds 50% / 30% / 20% in the top three ETFs.
- A slot whose ETF has a negative trailing return is held in cash; if all six are negative, the sub-strategy holds cash for 3 months.
- Results exclude trading costs and are benchmarked against SPY and QQQ buy-and-hold. The chart shows monthly weights per lookback.
- Average cash weight since 2011 ranged from 1.1% (24-month) to 26.6% (1-month).



Momentum Growth vs SPY and QQQ Since 2011

- $10,000 from January 2011 grew to $30,397 to $58,072 across the five lookbacks (before costs), vs $79,961 in SPY and $149,098 in QQQ.
- 6-month momentum grew the most ($58,072); 1-month the least ($30,397).
- No lookback matched SPY's or QQQ's ending value: QQQ's strong run made it a demanding benchmark.
- Momentum paths were smoother, with upside capture of 43.9% to 79.1% of SPY's across lookbacks.



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1-Month Momentum: Pros and Cons

- Pro: Shallower max drawdown since 2011 than SPY and QQQ (-22.8% vs -33.7% and -35.1%).
- Pro: Lower volatility than SPY (11.9% vs 17.0%), though its Sharpe was below SPY's and QQQ's, so the smoother path did not improve risk-adjusted return.
- Con: Worse than SPY since 2011 on CAGR (7.3% vs 14.1%, before costs).
- Con: Worse than SPY since 2011 on Sharpe (0.52 vs 0.77).
- Con: Highest turnover (105.9x); charged the MA strategy's costs, CAGR falls to 7.0%, below the MA strategy's 9.3%.
- Con: Often in cash: average cash weight of 26.6% and 42 full-cash months.

3-Month Momentum: Pros and Cons

- Pro: Shallower max drawdown since 2011 than SPY and QQQ (-22.4% vs -33.7% and -35.1%).
- Pro: Higher CAGR than the MA strategy after modeled costs (10.4% vs 9.3%); its Sharpe was in line with the MA strategy's (0.71 vs 0.67).
- Con: Worse than SPY since 2011 on CAGR (10.6% vs 14.1%, before costs).
- Con: Worse than SPY since 2011 on Sharpe (0.71 vs 0.77), and below QQQ's 0.86, despite lower volatility (13.1% vs 17.0%).
- Con: High turnover (60.7x), and weak years such as 2015 (-11.8% vs SPY 1.2%).

6-Month Momentum: Pros and Cons

- Pro: Shallowest max drawdown of all lookbacks since 2011 (-20.8% vs SPY -33.7% and QQQ -35.1%).
- Pro: Highest momentum CAGR, above the MA strategy after modeled costs (11.6% vs 9.3%), with a higher Sharpe than the MA strategy (0.79 vs 0.67).
- Con: Worse than SPY since 2011 on CAGR (11.8% vs 14.1%, before costs), and well below QQQ (18.7%).
- Con: Sharpe was only in line with SPY's (0.79 vs 0.77) and below QQQ's (0.86), so its smoother path did not add a clear risk-adjusted edge.
- Con: Turnover of 47.6x; weak years included 2015 (-8.2% vs SPY 1.2%) and 2019 (1.1% vs SPY 31.2%).

12-Month Momentum: Pros and Cons

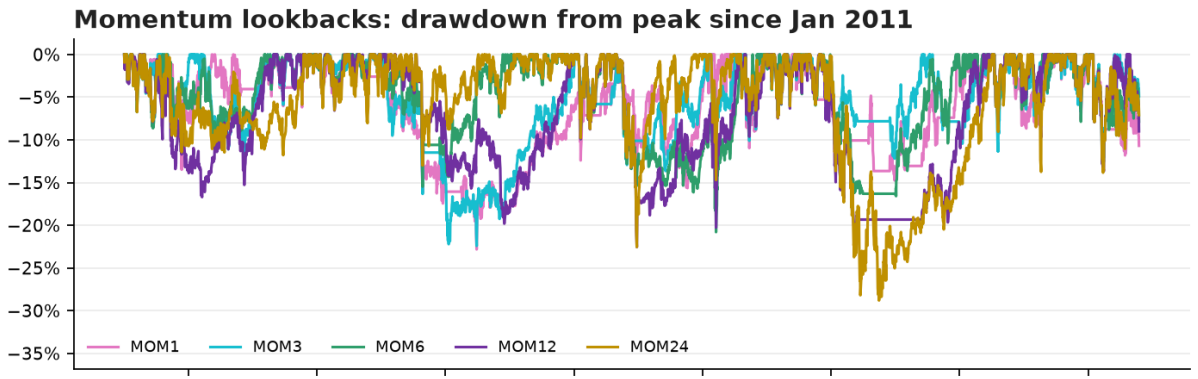
- Pro: Shallower max drawdown since 2011 than SPY and QQQ (-22.5% vs -33.7% and -35.1%).
- Pro: Lower turnover than the shorter lookbacks (34.3x) and a strong 2025 (41.0% vs SPY 17.7% and QQQ 20.8%).
- Con: Worse than SPY since 2011 on CAGR (9.0% vs 14.1%, before costs); after modeled costs it was in line with the MA strategy (8.9% vs 9.3%).
- Con: Worse than SPY since 2011 on Sharpe (0.58 vs 0.77), and below the MA strategy's 0.67.
- Con: Volatile year-to-year results, e.g. 2016 (-10.6% vs SPY 12.0%).

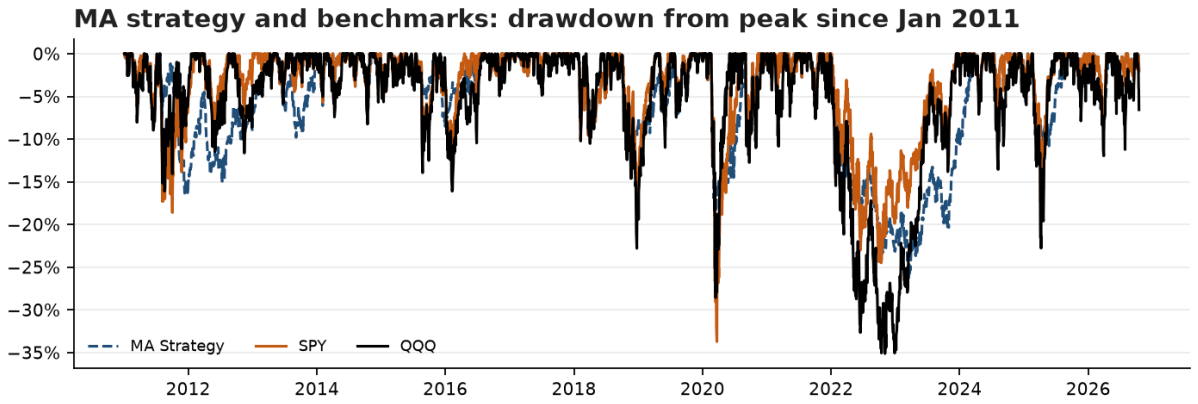
24-Month Momentum: Pros and Cons

- Pro: Shallower max drawdown since 2011 than SPY and QQQ (-28.8% vs -33.7% and -35.1%).
- Pro: Lowest turnover of the lookbacks (23.8x) and the highest upside capture (79.1% of SPY's).
- Con: Worse than SPY since 2011 on CAGR (10.3% vs 14.1%, before costs); after modeled costs it was in line with the MA strategy (10.2% vs 9.3%).
- Con: Worse than SPY since 2011 on Sharpe (0.62 vs 0.77), and below QQQ's 0.86.
- Con: Deepest drawdown of the lookbacks (-28.8%, vs the MA strategy's -26.4%); in 2022 it returned -23.1% vs SPY -18.2%.

Momentum Drawdowns by Lookback vs SPY and QQQ

- Max drawdowns since 2011: -20.8% (6-month), -22.4% (3-month), -22.5% (12-month), -22.8% (1-month) and -28.8% (24-month).
- Benchmarks: SPY -33.7% and QQQ -35.1%; the MA strategy -26.4%.
- In 2022, every lookback lost less than QQQ (-32.6%), but 24-month momentum (-23.1%) lost more than SPY (-18.2%).
- Shorter lookbacks used cash more often, which helped in sell-offs but cost return in recoveries.

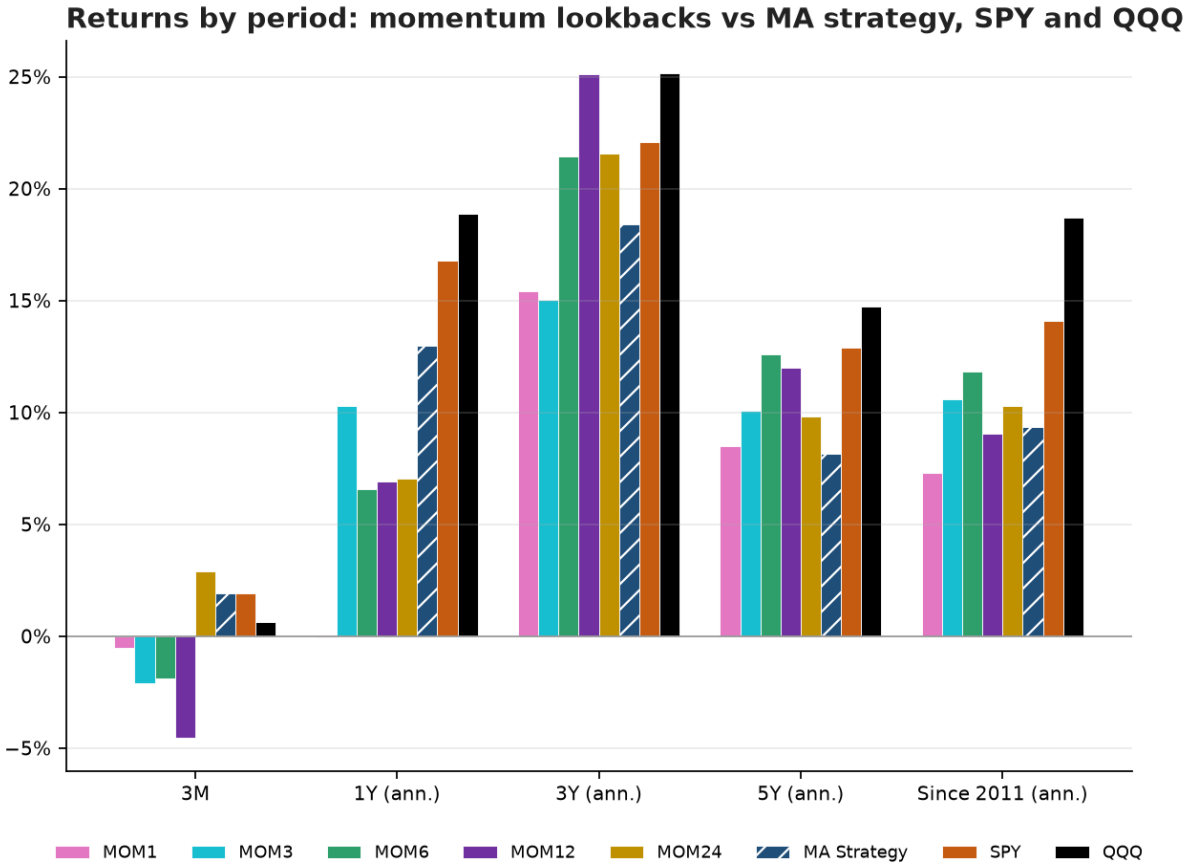




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Comparison Since 2011: Momentum vs MA Strategy vs SPY and QQQ

- CAGR: QQQ 18.7% and SPY 14.1% led; momentum ranged from 7.0% to 11.6% after modeled costs (7.3% to 11.8% before) vs the MA strategy's 9.3% net of costs.
- Sharpe: QQQ 0.86, 6-month momentum 0.79 and SPY 0.77 led; the MA strategy was 0.67 and the other lookbacks 0.52 to 0.71.
- Max drawdown: all momentum lookbacks (-20.8% to -28.8%) and the MA strategy (-26.4%) were shallower than SPY (-33.7%) and QQQ (-35.1%).
- Momentum results exclude trading costs; the MA strategy is net of modeled costs. Charged the MA strategy's costs, momentum CAGRs fall by small amounts.
- Best by measure: QQQ on CAGR and Sharpe; 6-month momentum on max drawdown and on CAGR and Sharpe among the strategies.



Head to Head Since 2011: Momentum vs MA Strategy vs SPY and QQQ

Return (3M total, else annualized)	3M	1Y	3Y	5Y	Since 2011
1-month momentum	-0.5%	0.0%	15.4%	8.5%	7.3%
3-month momentum	-2.1%	10.3%	15.0%	10.1%	10.6%
6-month momentum	-1.9%	6.6%	21.4%	12.6%	11.8%
12-month momentum	-4.5%	6.9%	25.1%	12.0%	9.0%
24-month momentum	2.9%	7.1%	21.6%	9.8%	10.3%
MA trend strategy	1.9%	13.0%	18.4%	8.1%	9.3%
SPY (buy and hold)	1.9%	16.8%	22.1%	12.9%	14.1%
QQQ (buy and hold)	0.7%	18.9%	25.1%	14.7%	18.7%

Since 2011	CAGR	Total	Vol.	Sharpe	Sortino	Max DD	Calmar	Beta	Corr.	Up cap.	Down cap.
1-month momentum	7.3%	204.0%	11.9%	0.52	0.72	-22.8%	0.32	0.28	0.39	43.9%	42.3%
3-month momentum	10.6%	387.7%	13.1%	0.71	0.99	-22.4%	0.47	0.31	0.40	53.0%	49.2%
6-month momentum	11.8%	480.7%	13.3%	0.79	1.10	-20.8%	0.57	0.41	0.52	58.7%	54.4%
12-month momentum	9.0%	290.7%	13.8%	0.58	0.81	-22.5%	0.40	0.49	0.60	62.7%	62.2%
24-month momentum	10.3%	368.3%	15.1%	0.62	0.86	-28.8%	0.36	0.66	0.74	79.1%	79.8%
MA trend strategy	9.3%	307.5%	12.1%	0.67	0.91	-26.4%	0.35	0.41	0.58	65.3%	65.1%
SPY (buy and hold)	14.1%	699.6%	17.0%	0.77	1.08	-33.7%	0.42	1.00	1.00	100.0%	100.0%
QQQ (buy and hold)	18.7%	1391.0%	20.7%	0.86	1.22	-35.1%	0.53	1.13	0.93	118.1%	115.5%

Common period 2011-01-03 to 2026-10-16. Momentum: 50% / 30% / 20% in the top three trailing performers of SPY, QQQ, TLT, GLD, IWM and IEF each month (negative slots in cash at 0%; 3 months in cash when all six are negative). **Momentum excludes trading costs; the MA strategy is net of modeled costs.** Ratios use the 13-week T-bill as the risk-free rate. Beta, correlation and capture are measured vs SPY.

Portfolio Fit: Who These Strategies May Suit

- Investors who prioritise shallower drawdowns and accept lower long-run returns than SPY or QQQ in exchange.
- As a complement to a core equity allocation: since 2007 the MA strategy's correlation to SPY was 0.45; since 2011 momentum correlations ranged from 0.39 to 0.74.
- Momentum diversifies across equities, Treasuries and gold; the MA strategy is a simpler two-asset SPY/IEF switch.
- Less suited to investors focused on maximising return or matching an equity benchmark year to year.
- Both rely on trends persisting; choppy markets and joint stock-bond declines, as in 2022, have been the main weak spots.

Next Steps

- Request the full methodology document, including signal definitions and data sources.
- Request the trading-cost model and execution assumptions used for the MA strategy.
- Discuss sizing and implementation within your existing portfolio and constraints.
- Request ongoing monthly reporting of allocations and performance.

For discussion purposes only. Not an offer or solicitation.

Momentum Detail: Calendar-Year Returns and Growth of $10,000

Year	1-month	3-month	6-month	12-month	24-month	MA	SPY	QQQ
2011	2.5%	23.5%	12.9%	-5.1%	-2.6%	-8.3%	1.9%	3.5%
2012	9.0%	5.6%	-2.0%	3.6%	2.3%	8.0%	16.0%	18.1%
2013	10.2%	29.4%	35.9%	33.2%	15.3%	17.4%	32.3%	36.6%
2014	0.1%	12.1%	14.0%	14.0%	10.5%	12.0%	13.5%	19.2%
2015	-11.3%	-11.8%	-8.2%	-3.5%	3.9%	-4.5%	1.2%	9.4%
2016	-2.0%	-2.9%	14.4%	-10.6%	2.2%	13.6%	12.0%	7.1%
2017	13.4%	15.9%	17.1%	21.7%	24.9%	21.7%	21.7%	32.7%
2018	-3.4%	-2.9%	0.5%	-4.0%	-3.5%	1.6%	-4.6%	-0.1%
2019	5.2%	7.2%	1.1%	7.3%	25.6%	16.2%	31.2%	39.0%
2020	50.5%	33.5%	29.2%	21.0%	22.5%	12.9%	18.3%	48.4%
2021	9.1%	21.0%	23.2%	17.4%	24.5%	28.7%	28.7%	27.4%
2022	-9.3%	-8.5%	-15.5%	-16.2%	-23.1%	-22.7%	-18.2%	-32.6%
2023	16.2%	11.1%	25.2%	13.6%	13.5%	18.3%	26.2%	54.9%
2024	6.0%	9.9%	16.7%	20.4%	23.1%	24.9%	24.9%	25.6%
2025	32.8%	26.4%	30.1%	41.0%	32.7%	10.8%	17.7%	20.8%
2026 YTD	1.0%	9.4%	6.1%	5.5%	5.7%	9.1%	12.8%	15.9%

Growth of $10,000 since 2011	Value
1-month momentum	$30,397
3-month momentum	$48,772
6-month momentum	$58,072
12-month momentum	$39,068
24-month momentum	$46,830
MA trend strategy	$40,753
SPY (buy and hold)	$79,961
QQQ (buy and hold)	$149,098

Calendar-year returns are total returns; the final year is year-to-date through 2026-10-16. Momentum excludes trading costs; MA = moving-average trend strategy, net of modeled costs.

Momentum Detail: Allocations, Activity and Cost Sensitivity

Average weight since 2011	SPY	QQQ	TLT	GLD	IWM	IEF	CASH
1-month momentum	13.1%	19.1%	9.9%	12.1%	15.0%	4.2%	26.6%
3-month momentum	16.1%	25.3%	11.1%	15.4%	16.9%	3.5%	11.8%
6-month momentum	17.4%	27.7%	10.7%	15.1%	18.1%	4.3%	6.7%
12-month momentum	16.9%	30.9%	10.1%	15.1%	16.2%	4.0%	6.9%
24-month momentum	22.1%	38.1%	8.8%	14.6%	14.5%	0.8%	1.1%

Current weights (2026-10)	SPY	QQQ	TLT	GLD	IWM	IEF	CASH
1-month momentum	-	50.0%	-	-	-	-	50.0%
3-month momentum	30.0%	20.0%	-	50.0%	-	-	-
6-month momentum	30.0%	50.0%	-	-	20.0%	-	-
12-month momentum	20.0%	50.0%	-	-	30.0%	-	-
24-month momentum	20.0%	30.0%	-	50.0%	-	-	-

Activity since 2011	Months	Rebalances	All-cash months	Avg cash weight	Turnover*	CAGR as shown	CAGR at 5 bps
1-month momentum	190	156	42	26.6%	105.9x	7.3%	7.0%
3-month momentum	190	139	18	11.8%	60.7x	10.6%	10.4%
6-month momentum	190	136	9	6.7%	47.6x	11.8%	11.6%
12-month momentum	190	112	12	6.9%	34.3x	9.0%	8.9%
24-month momentum	190	93	0	1.1%	23.8x	10.3%	10.2%

*Turnover = how many times the whole portfolio was traded since 2011 (e.g. 50.0x) (sum of monthly target-weight changes, halved; the initial purchase counts as 1.0). "CAGR at 5 bps" charges that cost per unit of turnover; momentum results elsewhere exclude trading costs. Weights are monthly targets; holdings drift within each month.

Meet the Digital Investment Team (AI-generated, fictional)

Margaret Okonkwo-Hale

Lead Portfolio Manager, Systematic Allocation

Focus: Overall strategy oversight, portfolio construction and allocation decisions

Certifications: CFA, CAIA | **Experience:** 26 yrs

Education

BA, Economics, University of Michigan (2000)
MBA, Finance, University of Chicago Booth School of Business (2006)

Experience

Analyst, Asset Allocation, Harborview Capital Partners (2000-2006)
Senior Portfolio Manager, Stonebridge Quantitative Advisors (2006-2015)
Head of Systematic Strategies, Ridgeline Tactical Investments (2015-2026)

Margaret Okonkwo-Hale began her career in asset allocation research before spending more than a decade running rules-based multi-asset portfolios. On this team she owns final responsibility for both the SPY/IEF trend rotation and the multi-asset momentum strategy, approving any change to rules, rebalancing schedules or the asset universe. She chairs the monthly review where signals, risk and execution are discussed together. Her distinctive strength is discipline: she insists that every rule be written down, tested out of sample and followed consistently, resisting the temptation to override systematic signals during volatile markets.

AI-generated fictional persona

Daniel Reyes-Whitford

Senior Quantitative Analyst

Focus: Quantitative research and signal design (moving-average and momentum signals)

Certifications: CQF | **Experience:** 12 yrs

Education

BS, Mathematics, University of Texas at Austin (2009)
PhD, Statistics, Carnegie Mellon University (2014)

Experience

Quantitative Researcher, Lattice Point Research (2014-2019)
Senior Quantitative Analyst, Ridgeline Tactical Investments (2019-2026)

Daniel Reyes-Whitford holds a doctorate in statistics and spent five years building time-series models at a quantitative research boutique. He owns signal design for both strategies, including the moving-average crossover logic that drives the SPY/IEF rotation and the lookback and ranking methodology behind the monthly momentum model. He documents every parameter choice and its rationale. His distinctive strength is skepticism toward overfitting: he routinely stress-tests signals across alternative lookback windows and subperiods to confirm that results reflect robust behavior rather than lucky parameter selection.

AI-generated fictional persona

Priya Natarajan

Risk Manager

Focus: Risk management, drawdown monitoring and position limits

Certifications: FRM, PRM | **Experience:** 16 yrs

Education

BCom, Finance, University of Toronto (2008)
MS, Financial Engineering, Columbia University (2010)

Experience

Risk Analyst, Northgate Risk Analytics (2010-2016)
Senior Risk Associate, Cobalt Ridge Asset Management (2016-2021)
Risk Manager, Ridgeline Tactical Investments (2021-2026)

Priya Natarajan trained in financial engineering and spent over a decade in independent risk oversight roles. On this team she owns risk measurement for both strategies, tracking volatility, drawdowns, correlation shifts between equities and Treasuries, and concentration when momentum favors a single asset. She reports independently to the portfolio manager and maintains the stress-scenario library. Her distinctive strength is scenario thinking: she builds historical and hypothetical stress cases, such as simultaneous equity and bond selloffs, to show investors how the strategies may behave when diversification assumptions break down.

AI-generated fictional persona

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Digital Investment Team, continued (AI-generated, fictional)

Thomas Brennan-Achebe

Macro and Rates Analyst

Focus: Macro and interest rates, including IEF and TLT duration exposure

Certifications: CFA | **Experience:** 11 yrs

Education

BA, Economics, Georgetown University (2012)
MA, International Economics, Johns Hopkins University SAIS (2015)

Experience

Rates Research Associate, Meridian Fixed Income Advisors (2015-2020)
Macro and Rates Analyst, Ridgeline Tactical Investments (2020-2026)

Thomas Brennan-Achebe started in fixed income research, covering Treasury curve dynamics and central bank policy. He owns the macro and rates perspective on this team, explaining how the IEF defensive leg of the trend rotation and the TLT and IEF holdings in the momentum strategy respond to inflation, policy and duration risk. He provides context rather than overrides to the systematic signals. His distinctive strength is translating rate-regime shifts into plain language, helping investors understand why Treasuries may not always act as a safe haven.

AI-generated fictional persona

Kenji Morales

Data Engineering Analyst

Focus: Market data engineering and backtest integrity

Certifications: None | **Experience:** 7 yrs

Education

BS, Computer Science, Georgia Institute of Technology (2019)

Experience

Data Engineer, Clearwater Signal Labs (2019-2023)
Data Engineering Analyst, Ridgeline Tactical Investments (2023-2026)

Kenji Morales is a computer scientist who built market data pipelines at a fintech startup before joining the team. He owns the data infrastructure behind both strategies, including adjusted price histories for SPY, QQQ, TLT, GLD, IWM and IEF, dividend handling and reproducible backtest code. He maintains version control for every research run. His distinctive strength is backtest hygiene: he systematically checks for look-ahead bias, survivorship issues and misaligned rebalance dates, ensuring that research results reflect information that would genuinely have been available at the time.

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Sofia Lindqvist-Barros

Trading and Investor Communications Analyst

Focus: Trading, execution and investor communications

Certifications: CMT, CIPM | **Experience:** 8 yrs

Education

BBA, Finance, Boston College (2017)
MS, Finance, Stockholm School of Economics (2018)

Experience

Trading Associate, Bayshore Execution Services (2018-2022)
Trading and Communications Analyst, Ridgeline Tactical Investments (2022-2026)

Sofia Lindqvist-Barros began her career on an ETF execution desk, where she learned to manage spreads, liquidity windows and order timing. She owns trade implementation for both strategies' rebalances and prepares investor reporting that explains each month's positioning. She bridges the technical team and clients, turning rules and risk reports into clear updates. Her distinctive strength is clarity: she writes communications that describe what the strategies do, why they changed position and what risks remain, without hype or unsupported promises.

AI-generated fictional persona

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Supporting Data: Calendar Years and Rolling Returns

Year	Strategy	SPY	60/40
2007	-0.7%	5.1%	7.7%
2008	-6.8%	-36.8%	-16.7%
2009	14.7%	26.4%	13.3%
2010	-1.8%	15.1%	13.4%
2011	-8.3%	1.9%	8.3%
2012	8.0%	16.0%	11.3%
2013	17.4%	32.3%	15.6%
2014	12.0%	13.5%	12.0%
2015	-4.5%	1.2%	1.8%
2016	13.6%	12.0%	7.8%
2017	21.7%	21.7%	13.8%
2018	1.6%	-4.6%	-2.0%
2019	16.2%	31.2%	21.8%
2020	12.9%	18.3%	16.9%
2021	28.7%	28.7%	15.1%
2022	-22.7%	-18.2%	-16.4%
2023	18.3%	26.2%	17.0%
2024	24.9%	24.9%	14.2%
2025	10.8%	17.7%	14.3%
2026 YTD	9.1%	12.8%	6.3%

Rolling 1-year returns	Strategy	SPY
Best 1-year period	44.6%	77.5%
Worst 1-year period	-23.1%	-47.4%
Periods beating SPY	18.7%	
Median excess vs SPY	-4.1%	
Growth of $10,000 (since walk-forward start)		
Strategy	$42,495	
SPY	$77,253	
60/40	$48,905	

Calendar-year returns are total returns. The final year is year-to-date through 2026-10-16. Rolling figures cover every 252-trading-day window since the walk-forward start. 60/40 is rebalanced daily, without costs.

Important Disclosures

Hypothetical performance. All performance shown is hypothetical and backtested. It does not represent actual trading or actual client accounts. Backtested results are prepared with the benefit of hindsight and have inherent limitations. No representation is made that any account will or is likely to achieve similar results.

Methodology. The strategy holds either SPY or IEF, based on whether SPY closed above a moving average the previous day. The moving average (5, 20, 50 or 200 days) is chosen each January using only data available before that date (walk-forward). Results include a modeled trading cost of 5 basis points per full switch between SPY and IEF. They exclude management fees, taxes and market impact beyond that assumption, and use adjusted closing prices, which include dividends. Partial calendar years are year-to-date.

Momentum methodology. Five monthly momentum sub-strategies (1, 3, 6, 12 and 24-month lookbacks) rank SPY, QQQ, TLT, GLD, IWM and IEF by trailing return at each month-end, and hold 50%, 30% and 20% in the top three for the following month, from January 2011, rebalanced monthly. A slot whose ETF has a negative trailing return is held in cash at an assumed 0% return; if all six are negative, the whole sub-strategy holds cash for three months and then resumes. QQQ and SPY buy-and-hold are shown as benchmarks. Momentum results **do not include trading costs**, while the moving-average strategy is shown net of modeled costs, so comparisons between them favor momentum by the cost of its switches. Cash earning 0% understates what T-bills paid in some periods.

In-sample results. Comparisons of individual moving-average rules over the full period are in-sample. They are shown for context only, and overstate what an investor could have known in advance.

Risk. Investing involves risk, including loss of principal. Trend-following signals can lag reversals, cause frequent switching in choppy markets, and underperform buy-and-hold for long periods. Treasury ETFs carry interest-rate risk and can fall alongside equities.

Data. Prices from Yahoo Finance via the yfinance library. The data has not been independently verified. The risk-free rate is the 13-week Treasury bill yield (^IRX).

Investment team. The investment team members in this presentation are AI-generated fictional personas created by an AI system for illustration. They are not real people, and their stated employers are fictional.

Not advice. This presentation was produced by an automated multi-agent research prototype. It is for informational purposes only, and is not investment advice, an offer, or a solicitation to buy or sell any security.